

REPORT

ARTIFICIAL INTELLIGENCE (CSA1717)

Assessment Tool 2 – Scenario Based Assignment

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1.INTRODUCTION

Artificial Intelligence (AI) has become an essential technology for solving complex real-world problems across healthcare, transportation, space exploration, education, and gaming. AI techniques such as search algorithms, optimization methods, Constraint Satisfaction Problems (CSP), and game-playing algorithms enable intelligent systems to make accurate decisions in dynamic environments. These techniques improve efficiency, reduce computational complexity, and support real-time decision-making.

2.AI-POWERED DRONE FOR EMERGENCY MEDICINE DELIVERY

2.1 PROBLEM STATEMENT

Develop an AI-based drone navigation system that delivers emergency medicines in flood-affected regions. The system must determine the safest and shortest route despite blocked roads, changing weather conditions, and varying travel costs. The solution should minimize delivery time while ensuring safe and efficient navigation.

2.3 OBJECTIVE

- Deliver emergency medicines quickly and safely.
- Find the shortest and safest route.
- Adapt to blocked roads and changing weather conditions.
- Reduce delivery time and operational risk.
- Improve emergency response efficiency.

2.4 ALGORITHM USED

A* Search Algorithm

2.5 SOLUTION

The proposed solution uses the A* Search algorithm to determine the safest and shortest route for medicine delivery. The algorithm calculates the optimal path using both the actual travel cost and a heuristic estimate of the remaining distance. When obstacles such as blocked roads or adverse weather are detected, the drone updates its route in real time. This approach ensures fast, reliable, and safe medicine delivery even in dynamic disaster environments.

2.6 ADVANTAGES

- Finds the optimal route efficiently.
- Supports real-time route replanning.
- Reduces delivery time.
- Avoids blocked or unsafe paths.
- Improves emergency response and patient safety.

2.7 CONCLUSION

An AI-powered drone navigation system using the A* Search algorithm provides an effective solution for emergency medicine delivery in disaster-affected areas. By combining intelligent path planning with real-time environmental updates, the system ensures safe, reliable, and timely delivery of medicines, helping save lives during emergencies.

3. SMART CITY TRAFFIC SIGNAL OPTIMIZATION

3.1 INTRODUCTION

Traffic congestion is a major problem in modern cities, leading to increased travel time, fuel consumption, and air pollution. Traditional traffic signal systems use fixed timings, which cannot adapt to changing traffic conditions. Artificial Intelligence (AI) provides an intelligent solution by optimizing traffic signal timings based on real-time traffic flow. This improves traffic movement, reduces waiting time, and enhances transportation efficiency.

3.2 PROBLEM STATEMENT

Develop an AI-based traffic signal optimization system that minimizes traffic congestion, vehicle waiting time, and fuel consumption by dynamically adjusting traffic signal timings according to real-time traffic conditions. The system should efficiently handle a large search space and avoid getting trapped in locally optimal solutions.

3.3 OBJECTIVES

- Reduce traffic congestion.
- Minimize vehicle waiting time.
- Reduce fuel consumption.

- Optimize traffic signal timings.
- Improve overall traffic flow.

3.4 EXISTING SYSTEM

- Uses fixed traffic signal timings.
- Cannot adapt to changing traffic conditions.
- Causes long waiting times.
- Increases fuel consumption and pollution.

3.5 PROPOSED SYSTEM

- Uses AI to optimize traffic signals.
- Adjusts signal timings dynamically.
- Responds to real-time traffic flow.
- Reduces congestion and delays.
- Improves road efficiency.

3.6 ALGORITHM USED

Genetic Algorithm (GA)

3.7 WORKING METHODOLOGY

- Collect real-time traffic data.
- Generate different traffic signal timing plans.
- Evaluate each plan based on traffic performance.
- Apply selection, crossover, and mutation.
- Choose the best signal timing.
- Continuously update timings as traffic changes.

3.8 SOLUTION

The proposed solution uses a Genetic Algorithm to optimize traffic signal timings. The algorithm creates multiple signal timing combinations and evaluates them using factors such as waiting time, traffic congestion, and fuel consumption. Through selection, crossover, and mutation, it continuously improves the solutions and adapts to changing traffic conditions, providing efficient traffic management.

3.9 ADVANTAGES

- Reduces traffic congestion.
- Minimizes waiting time.
- Saves fuel.
- Adapts to real-time traffic.
- Provides scalable traffic management.

3.10 APPLICATIONS

- Smart cities.
- Urban traffic management.
- Highway traffic control.
- Intelligent Transportation Systems (ITS).
- Emergency vehicle signal management.

3.11 RESULT

The AI-based traffic signal optimization system successfully reduces congestion, decreases vehicle waiting time, and improves traffic flow by dynamically adjusting signal timings based on real-time traffic conditions.

3.12 CONCLUSION

The Genetic Algorithm provides an effective solution for smart city traffic management by optimizing traffic signal timings in real time. It reduces congestion, improves traffic flow, minimizes fuel consumption, and supports intelligent transportation systems for modern cities.

4. AUTONOMOUS MARS ROVER NAVIGATION

4.1 PROBLEM STATEMENT

Develop an AI-based autonomous rover capable of exploring unknown Martian terrain, collecting samples, avoiding obstacles, and making real-time navigation decisions with limited sensor information. The rover should continuously update its knowledge of the environment to ensure safe and efficient exploration.

4.2 OBJECTIVES

- Explore unknown terrain safely.
- Avoid rocks, craters, and other hazards.
- Collect soil and rock samples efficiently.
- Make real-time navigation decisions.
- Continuously update the rover's internal map.

4.3 EXISTING SYSTEM

- Relies on predefined maps.
- Limited ability to adapt to unknown environments.
- Requires frequent human intervention.
- Less effective in dynamic conditions.

4.4 PROPOSED SYSTEM

- Uses an **Online Search Agent** for navigation.
- Collects real-time data from sensors and cameras.
- Builds and updates an internal map during exploration.
- Detects obstacles and replans routes automatically.
- Navigates safely with minimal human intervention.

4.5 ALGORITHM USED

Online Search Agent

4.6 WORKING METHODOLOGY

1. Start the rover mission.
2. Collect data using onboard sensors and cameras.
3. Detect obstacles and safe paths.
4. Update the internal map continuously.
5. Select the best route to the destination.
6. Replan the path if new obstacles are found.
7. Reach the target and collect samples.

4.7 SOLUTION

The proposed solution uses an Online Search Agent that enables the rover to make decisions while exploring unknown terrain. It continuously gathers information from sensors, updates its internal map, detects hazards, and replans routes whenever new obstacles are found. This approach ensures safe navigation, efficient exploration, and successful sample collection even in uncertain environments.

4.8 ADVANTAGES

- Supports real-time decision-making.
- Safely avoids obstacles.
- Adapts to unknown environments.
- Improves exploration efficiency.
- Reduces human intervention.

4.9 APPLICATIONS

- Mars and Moon exploration.
- Space research missions.
- Search and rescue robots.
- Military surveillance.
- Autonomous vehicles.

4.10 RESULT

The rover successfully navigates unknown terrain, avoids hazards, updates its map in real time, and safely reaches its destination while collecting samples.

4.11 CONCLUSION

The provides an effective solution for autonomous Mars rover navigation. It enables the rover to adapt to changing environments, make intelligent decisions, and complete exploration missions safely and efficiently.

5.UNIVERSITY EXAM TIMETABLE GENERATION USING CONSTRAINT SATISFACTION PROBLEM (CSP)

5.1 PROBLEM STATEMENT

Develop an AI-based university exam timetable generation system that schedules exams without conflicts while satisfying multiple constraints such as limited exam halls, faculty availability, subject precedence, and special accommodations for students. The system should efficiently generate a valid timetable that is scalable for a large number of courses and students.

5.2 OBJECTIVES

- Generate a conflict-free exam timetable.
- Ensure no student has overlapping exams.
- Efficiently allocate exam halls and faculty.
- Satisfy all scheduling constraints.
- Produce a scalable timetable for large universities.

5.3 EXISTING SYSTEM

- Manual timetable preparation is time-consuming.
- Scheduling conflicts may occur.
- Difficult to manage multiple constraints.
- Requires frequent modifications and human effort.

5.4 PROPOSED SYSTEM

- Uses **Constraint Satisfaction Problem (CSP)** techniques.
- Assigns exams to suitable time slots and halls.
- Applies constraints to avoid scheduling conflicts.
- Uses Backtracking and Forward Checking for efficient scheduling.
- Automatically generates an optimized exam timetable.

5.5 ALGORITHM USED

Constraint Satisfaction Problem (CSP) with Backtracking and Forward Checking

5.6 WORKING METHODOLOGY

- Define exams as variables.
- Assign available time slots and halls as domains.
- Apply scheduling constraints.
- Use Backtracking Search to assign valid slots.
- Apply Forward Checking to remove conflicting choices.
- Generate the final conflict-free timetable.

5.7 SOLUTION

The proposed solution models the exam scheduling problem as a Constraint Satisfaction Problem (CSP). Each exam is represented as a variable, while available time slots and exam halls form the domains. Constraints such as avoiding overlapping exams, faculty availability, hall capacity, subject order, and special accommodations are enforced during scheduling. Backtracking Search combined with Forward Checking efficiently generates a valid and conflict-free timetable by eliminating invalid assignments early in the search process.

5.8 ADVANTAGES

- Generates conflict-free schedules.
- Reduces manual effort.
- Handles multiple constraints efficiently.
- Saves time in timetable preparation.
- Easily scalable for large universities.

5.9 APPLICATIONS

- Universities and colleges.
- School examination scheduling.
- Training and certification programs.
- Competitive examination scheduling.
- Academic timetable management systems.

5.10 RESULT

The AI-based CSP system successfully generates a valid exam timetable by satisfying all scheduling constraints, reducing conflicts, and improving the efficiency of timetable preparation.

5.11 CONCLUSION

The Constraint Satisfaction Problem (CSP) with Backtracking and Forward Checking provides an efficient solution for university exam timetable generation. It ensures conflict-free scheduling, satisfies multiple constraints, reduces manual effort, and supports scalable timetable management for educational institutions.

6. STRATEGIC GAME AI FOR REAL- TIME DECISION MAKING USING MINIMAX AND ALPHA-BETA PRUNING

6.1 PROBLEM STATEMENT

Develop an AI-based game-playing system for a real-time strategy game that makes intelligent decisions within strict time limits. The system must evaluate multiple possible game states, efficiently search a large decision tree using the Minimax algorithm with Alpha-Beta Pruning, and select the best move while maintaining optimal performance in real time.

6.2 OBJECTIVES

- Develop an intelligent AI opponent.
- Make optimal decisions in real time.
- Reduce computation time during gameplay.
- Improve game performance and strategy.
- Efficiently evaluate large game trees.

6.3 EXISTING SYSTEM

- Uses basic rule-based or random decision-making.
- Evaluates many unnecessary game states.
- Requires more computation time.
- Performs poorly in complex game situations.
- May fail to make optimal decisions within time limits.

6.4 PROPOSED SYSTEM

- Uses the Minimax algorithm for intelligent decision-making.
- Applies Alpha-Beta Pruning to eliminate unnecessary branches.
- Evaluates possible future game states efficiently.
- Selects the best move within the given time limit.
- Improves gameplay performance and responsiveness.

6.5 ALGORITHM USED

Minimax Algorithm with Alpha-Beta Pruning

6.6 WORKING METHODOLOGY

- Represent the game as a decision tree.
- Generate all possible moves.
- Evaluate each game state using an evaluation function.
- Apply the Minimax algorithm to determine the best move.
- Use Alpha-Beta Pruning to remove unnecessary branches.
- Select and execute the optimal move.

6.7 SOLUTION

The proposed solution uses the Minimax algorithm with Alpha-Beta Pruning to develop an intelligent game-playing AI. Minimax evaluates future game states by considering both the AI's and the opponent's possible moves. Alpha-Beta Pruning improves efficiency by eliminating branches that cannot affect the final decision, significantly reduce computation time while maintain the same optimal result. This approach enables fast, accurate, and intelligent decision-making in real-time strategy games.

6.8 ADVANTAGES

- Makes intelligent game decisions.
- Reduces computation time.
- Improves real-time performance.
- Selects optimal moves efficiently.
- Handles complex game scenarios.

6.9 APPLICATIONS

- Chess AI.
- Tic-Tac-Toe AI.
- Checkers AI.
- Real-time strategy games.
- Board game and video game AI.

6.10 RESULT

The AI system successfully evaluates multiple game states, eliminates unnecessary computations using Alpha-Beta Pruning, and selects the optimal move within the required time limit, resulting in efficient and competitive gameplay.

6.11 CONCLUSION

The Minimax algorithm with Alpha-Beta Pruning provides an efficient solution for real-time game AI. It reduces computation time, improves decision-making speed, and enables the AI to make optimal moves while maintaining high performance in complex game environments.

7.CONCLUSION

Artificial Intelligence provides effective solutions for solving complex real-world problems through intelligent search, optimization, and decision-making techniques. This report presented five AI-based scenarios: Emergency Medicine Delivery using A* Search, Smart City Traffic Signal Optimization using Genetic Algorithm, Autonomous Mars Rover Navigation using an Online Search Agent, University Exam Timetable Generation using Constraint Satisfaction Problem (CSP), and Strategic Game AI using Minimax with Alpha-Beta Pruning. Each

algorithm was selected based on its ability to address the specific challenges of the given scenario.

The proposed solutions improve efficiency, reduce computation time, adapt to dynamic environments, and support optimal decision-making. These AI techniques demonstrate how intelligent systems can enhance healthcare, transportation, space exploration, education, and gaming applications. Overall, the report highlights the importance of AI algorithms in developing reliable, scalable, and real-time solutions for practical problems.

8.REFERENCES

- [1] S. Russell and P. Norvig, Artificial Intelligence: A Modern Approach, 4th ed. Pearson, 2021.
- [2] M. Wooldridge, Artificial Intelligence: Foundations of Computational Agents, 3rd ed. Cambridge University Press, 2021.
- [3] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2022 (reprint edition).
- [4] P. Flach, Simply Logical: Intelligent Reasoning by Example, 3rd ed. Cambridge University Press, 2023.
- [5] E. Rich, K. Knight, and S. B. Nair, Artificial Intelligence, McGraw-Hill Education, 2022.